👤 Donor Workflow (Blood / Organ Donor)

1. **Register / Login**
    - Chooses “Donor” role → enters personal, blood group, organ info → uploads ID proof.
    - Waits for hospital/admin verification.
2. **Verification**
    - Once verified, donor becomes visible to nearby recipients.
3. **View Nearby Requests**
    - Dashboard shows real-time donation requests that match donor’s blood type & location.
4. **Accept / Decline Request**
    - Donor accepts a matching request → backend updates request status → recipient & hospital get notified.
5. **Donation Process**
    - Hospital confirms appointment → donor donates → admin updates status to “Completed.”
6. **Post-Donation**
    - Cooldown period set (e.g., 90 days for blood).
    - Donation appears in donor’s history.
    - Auto-generated certificate available to download/share.
7. **Notifications**
    - Alerts for new requests, upcoming blood camps, or verification updates.

🩹 Recipient Workflow (Blood / Organ Requester)

1. **Register / Login**
    - Selects “Recipient” role → enters personal info, hospital details, and location.
2. **Create Request**
    - Fills form: blood/organ type, urgency, hospital name.
    - Backend saves and notifies nearby matching donors.
3. **Track Request Status**
    - Dashboard shows status:
      Pending → Matched → In Progress → Completed
4. **Donor Match**
    - When donor accepts, recipient gets real-time notification and donor details.
5. **Hospital Coordination**
    - Hospital confirms and schedules donation.
6. **Completion**
    - After donation, recipient marks request as completed or hospital updates it.
    - Option to rate donor or hospital (optional).

🏛️ Admin / Hospital Workflow

1. **Login (Admin Portal)**
    - Role-based access → Hospital or Super Admin.
2. **User Verification**
    - Reviews donor/recipient profiles & ID proofs.
    - Approves or rejects users (sets `verified = true`).

3. **Manage Requests**
    - Monitors all active donation requests.
    - Manually assigns donors if system doesn't auto-match.
    - Updates request status (In Progress / Completed).
4. **Organize Blood Camps / Events**
    - Creates new events with date, venue, capacity.
    - Visible on donor dashboards & map.
5. **Track Analytics**
    - Dashboard stats: total donations, top blood groups, active donors per city, etc.
6. **Moderation**
    - Blocks suspicious users, handles fake requests, ensures platform integrity.

🔄 End-to-End Flow Summary

Recipient → creates request
↓
Backend → notifies nearby verified donors
↓
Donor → accepts request
↓
Admin/Hospital → verifies & schedules donation
↓
Donation completed → status = "completed"
↓
Donor & Recipient notified + certificate issued

🩸 LIFELINK – Features Breakdown

👤 1. Donor Features

🔹 Registration & Profile

- Sign up / Login (Email, Phone, or Social Auth)
- Verify blood group, health status, and age (via medical form or hospital verification)
- View donor badge & donation history
- Location sharing (optional for nearby donation alerts)

🔹 Blood Donation Management

- Register availability to donate (toggle: Available/Unavailable)
- View nearby donation requests (filtered by location, blood type, urgency)
- Accept/Reject donation requests
- Schedule donation appointments

🔹 Rewards & Tracking

- Earn points or certificates for each successful donation
- Track past donations and recipient feedback
- Receive reminders for next eligible donation date

🔹 Notifications

- Push notifications for emergency requests nearby
- Donation appointment reminders
- Health checkup or campaign alerts

🧬 2. Recipient Features

🔹 **Registration & Profile**

- Sign up / Login (for patient, hospital staff, or family)
- Provide verified medical documents for request authenticity
- Add patient details (hospital name, blood group, urgency, quantity needed)

🔹 **Request Management**

- Create new blood/organ request
- View nearby matching donors
- Track donor responses and confirm appointment
- Option to mark request as “Urgent” or “Scheduled”

🔹 **Communication**

- View donor profile and rating
- Share hospital pickup details securely

🔹 **Notifications**

- Alerts when donor accepts request
- Status updates (pending, confirmed, completed)
- Admin-approved updates

👨‍💼 3. Admin Features

🔹 **Dashboard**

- Overview of total donors, recipients, and active requests
- Blood type availability and region-wise statistics
- Emergency alerts overview

🔹 **User Management**

- Approve/verify new donors and recipients
- Suspend or flag fake/inactive users
- Manage reports or issues from chat interactions

🔹 **Request Oversight**

- Monitor all active donation requests
- Auto-match donors based on proximity and blood type
- Manually intervene in emergencies

🔹 **Campaigns & Awareness**

- Create blood donation drives and awareness events
- Send push notifications or in-app banners
- Partner hospital onboarding

🔹 **Analytics**

- Track donation frequency and success rates
- Analyze region-wise demand-supply gap
- Generate monthly/weekly reports

🔐 4. Common / System Features

🔹 **Authentication & Security**

- OTP/email verification
- Encrypted data storage (HIPAA/GDPR compliant if applicable)
- Role-based access control (Donor, Recipient, Admin)

🔹 **Location & Matching**

- Geolocation to find nearby donors/recipients
- Smart matching algorithm (based on blood group, distance, availability, and urgency) (NEED TO FIGURE IT OUT)

🔹 **Other Enhancements**

- Multi-language support (English + local languages)
- Dark/light theme
- Feedback & rating system
- Donation certificates (auto-generated PDF)
- Leaderboard of top donors

💡 Optional Add-Ons (Future Scope)

- Integration with government/hospital databases
- Health tracking (connect with fitness bands or apps)
- Blood bank inventory module
- Gamification (badges, streaks, community goals)
- AI recommendation system (to predict shortage areas — for later)